MBA ZG583 · MANAGEMENT OF AI PRODUCTS · CAPSTONE

Executive Summary

UniConnect Operations OS — an AI-assisted academic operations platform

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1. Executive Summary

1.1 Problem

A higher-education network of 40 campuses has to run the same operational day, 40 times over, every working day. Fees have to be reconciled, daily campus reports have to be written and reviewed, faculty attendance has to be recorded, tasks have to be chased, budgets have to be approved, exams have to be planned, papers have to be generated, seating and invigilation have to be arranged, and problems have to reach the right person before they become escalations.

Until this platform existed, that work lived in spreadsheets, email threads, WhatsApp messages and disconnected Excel files. Three specific failures repeated every single day. First, information arrived late and in inconsistent formats, so a manager reviewing twelve campuses was reading twelve differently written reports. Second, the important signal was buried — a serious problem and a routine update looked identical in an inbox. Third, nothing was auditable: when a decision was questioned two weeks later, nobody could reconstruct who knew what, and when.

The platform solved the structure problem. What remains is a reading and judgement problem: the volume of daily operational text, numbers and documents now exceeds what a small central team can read carefully. That is the problem this capstone addresses.

1.2 Target User and Customer

The primary users are the people who run the operational day: Business Operations Associates (BOAs) and Programme Management Associates at each campus, who submit and act; Programme Managers and the Chief of Staff (COS), who review, sign off and escalate; and the central academic operations and product team, who see the network view. The customer, in the internal sense, is the operations leadership that funds the platform and carries the cost of delays, errors and escalations.

1.3 Product Concept

UniConnect Operations OS is a single role-scoped web platform that replaced the spreadsheet-and-email operating model with one auditable system: one database, one permission model, one record of what happened. It covers fee collection, the daily operations submission spine, tasks, communications and email campaigns, assessments and exam paper generation, examination classroom allocation, faculty and success-coach tracking, budget approvals, meeting intelligence, the operational event calendar and the academic operations hub.

On top of that platform sits the object of this capstone: the UniConnect Intelligence Layer. It is not a chatbot bolted onto a portal. It is a set of narrow, workflow-embedded AI capabilities that read the operational material nobody has time to read, decide how confident they are, and route each item to the right human with the evidence attached. The first wedge — the part recommended for pilot investment — is Daily Operations Triage: turning forty free-text daily campus reports plus their attached numbers into a ranked, evidence-linked review queue for the people who sign them off.

1.4 Why AI Is the Right Lever

Three properties of this workflow make AI the correct instrument rather than a fashionable one. The input is unstructured and written by forty different people in their own words, which defeats rule-based parsing. The judgement being made — "is this campus report normal, or does it hide something that needs attention today?" — is a repeated classification and ranking decision made hundreds of times a week. And the cost of the current approach is not a missing feature but human reading time, which does not fall as the network grows.

Rules, dashboards and thresholds already exist in the platform and they catch the numeric cases well: a fee number that drops, an attendance figure below target, a report that was not submitted by four o'clock. They cannot catch a sentence like "second-year students say the lab machines were down again since Monday" sitting in paragraph four of an otherwise routine report. That is a language problem, and language problems are where current models genuinely outperform rules.

There is a second kind of AI in this platform, and naming it matters because the course rightly warns against treating AI as a synonym for large language models. Allocating students to examination classrooms is a constrained combinatorial problem — every candidate seated, room capacities respected, and and no two neighbours drawn from the same branch or section — solved with search and repair rather than a language model. It is described in Section 9. Language AI reads and drafts; optimisation AI searches and places. Both are AI, they fail in completely different ways, and they need different evaluation. Equally important is where AI is not the right lever. The platform deliberately does not use AI for anything that must be exactly correct and legally defensible — fee arithmetic, permission enforcement, audit records and approval states are all deterministic code, and they should stay that way.

1.5 Recommendation

> Recommendation: PILOT. Run the Intelligence Layer on five campuses for one full academic month in assist-only mode, where every AI output is a suggestion attached to a human decision and no AI output is auto-sent or auto-approved. Continue to scale the deterministic platform across all 40 campuses in parallel. Do not extend AI into exam paper approval, student-facing communication or performance evaluation of staff until the pilot thresholds in Section 5.7 are met.

The reasoning behind a pilot rather than an immediate scale is that the binding risk here is not model quality but trust and workflow fit. If reviewers stop reading because the summary looked convincing, the platform has made oversight worse while appearing to make it faster. That risk is only measurable with real reviewers on real reports, which is exactly what a bounded pilot buys.